de Sitter at the FGHMV Standard: The First-Law Sign is Opposite, and the Isometries are Too Few

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: n4hy

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Abstract

Paper IX left cosmological extrapolation [**O**]. Paper 11 recorded a minus sign in the static-patch first law and did not attempt a derivation. This note *decides* Q2 at the only standard the program has used for a [**P**] gravitational theorem: Faulkner–Guica– Hartman–Myers–Van Raamsdonk.

That standard cannot be met in de Sitter by copying the AdS argument. Two identities are enough.

The Gibbons–Hawking entropy of four-dimensional de Sitter is $S = \pi\ell^2/G = 3\pi/(G\Lambda)$. Then $\partial S/\partial\Lambda < 0$: more vacuum energy, smaller horizon, less entropy [**?**]. The cosmological-horizon first law has a minus sign: adding Killing energy *decreases* horizon entropy [**?**]. The Casini–Huerta–Myers integrand is $+T_{00}$. Copying the AdS bookkeeping line to the cosmological horizon uses the wrong sign. [**P**]

Independently, $\dim \mathfrak{so}(1,4) = 10$ while the conformal algebra that supplies CHM special conformal maps in four dimensions is $\mathfrak{so}(2,4)$, of dimension 15. de Sitter isometries do not contain the CHM family of arbitrary balls. [**P**]

No dual is constructed. Jacobson is not used as a substitute. Linearized Einstein+Λ from a cosmological CFT remains unproved and is not labeled [**P**]. What *is* [**P**] is that FGHMV-standard closure of Q2 is obstructed.

The AdS results stand.

Standard of proof

"Close de Sitter" at this program's grade means the four-point checklist of Paper IX: a family of regions with controlled modular Hamiltonians; a first-law identity; equivalence to linearized Einstein–Cartan; Hehl–Datta from relative-entropy positivity.

Jacobson's entanglement equilibrium [**?**] is not that grade (Paper 14). Strominger's dS/CFT [**?**] is Euclidean and generically non-unitary, and is not a Maldacena pair. This note does not promote either.

A negative decision at the stated standard is a decision.

1 The AdS first-law sign

Casini–Huerta–Myers for a ball of radius R in a CFT vacuum is [?]

$$H_B = 2\pi \int_{|x|<R} d^{d-1}x \, \frac{R^2 - |x|^2}{2R} \, T_{00}(x). \tag{1}$$

The kernel of T_{00} is positive inside the ball. Faulkner et al. convert $\delta S_B = \delta\langle H_B\rangle$ into linearized Einstein equations about AdS with the standard energy sign [?]: a positive $\delta\langle T_{00}\rangle$ increases ball entanglement entropy. Relative-entropy positivity locks that direction.

2 The de Sitter identities

Four-dimensional de Sitter of radius ℓ has $\Lambda = 3/\ell^2$ and a cosmological horizon of area $A = 4\pi\ell^2$. The Gibbons–Hawking entropy is [?]

$$S_{\mathrm{GH}} = \frac{A}{4G} = \frac{\pi\ell^2}{G} = \frac{3\pi}{G\Lambda}. \tag{2}$$

Differentiating at fixed G,

$$\frac{\partial S_{\mathrm{GH}}}{\partial \Lambda} = -\frac{3\pi}{G\Lambda^2} < 0. \tag{3}$$

Increasing the vacuum energy decreases the horizon entropy. Banihashemi, Jacobson, Svesko and Visser isolate the same minus sign in the static-patch first law: adding Killing energy reduces cosmological-horizon entropy [?].

Theorem 2.1 (sign obstruction)**.** *The FGHMV first law and the Gibbons–Hawking static-patch first law have opposite signs for the response of entropy to Killing energy / vacuum energy. A line-by-line copy of the AdS bookkeeping onto the cosmological horizon is therefore false.* ***[P]***

A mutation confirms the derivative is not tautological. For AdS, $\Lambda = -3/\ell^2$ and the same area law $S = \pi\ell^2/G$ give $\partial S/\partial\Lambda > 0$ along that branch. The de Sitter sign is the one that conflicts with (**??**).

3 The isometry obstruction

The isometry algebra of dS_4 is $\mathfrak{so}(1,4)$, of dimension 10. The conformal algebra of four-dimensional Minkowski space, which supplies the special conformal transformations used to map an arbitrary ball to a Rindler wedge in the CHM construction, is $\mathfrak{so}(2,4)$, of dimension 15.

Theorem 3.1 (too few isometries)**.** $\dim \mathfrak{so}(1,4) = 10 < 15 = \dim \mathfrak{so}(2,4)$. *The de Sitter isometry group does not contain the CHM family of arbitrary-radius balls. The only regions with an exact Killing modular flow for a de Sitter-invariant state are static patches (and their complements), not a FGHMV net of balls.* ***[P]*** *(Lie-algebra dimension; the static-patch clause is the standard Killing-horizon statement, cited as such).*

de Sitter is conformally flat, so a *CFT living on a fixed* dS_4 background can still use $\mathfrak{so}(2,4)$. That is entanglement of matter *on* de Sitter, not a derivation of the Einstein equation with Λ as dynamics. Q2 is the latter. Conformal flatness does not supply a holographic pair, a timelike boundary dictionary, or a Ryu–Takayanagi theorem at the FGHMV grade.

4 What the small-ball limit does prove

For a ball of proper radius $R \ll \ell$ in a fixed de Sitter geometry, $g = \eta + O(R^2/\ell^2)$ and H_B is the Minkowski CHM plus $O(R^2/\ell^2)$. The first law in that limit is the Minkowski / AdS one. It does *not* derive $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ as an identity from a family of finite balls. The cosmological constant is invisible until $R \sim \ell$, which is exactly the regime in which Theorem **??** applies.

The monograph's estimate $\Lambda L^2 \ll 1$ below the Hubble scale [**?**] is unchanged and is not a dS derivation.

5 Machine check

Solver `m9_6_ds_sign.py`; auditor `m9_6_audit_ds.py` (sympy $S(\Lambda)$, combinatorial $\mathfrak{so}(n)$, no solver import). Gates C1–C5 PASS. The auditor recovers $S = 3\pi/(G\Lambda)$ and $\partial S/\partial\Lambda = -3\pi/(G\Lambda^2)$ identically, and counts $\dim \mathfrak{so}(5) = 10$, $\dim \mathfrak{so}(6) = 15$.

6 Verdict

Claim	Status
$S_{\mathrm{GH}} = 3\pi/(G\Lambda)$, $\partial S/\partial\Lambda < 0$	[**P**], audited
FGHMV / CHM sign is $+\delta\langle T_{00}\rangle$	[**P**] (cited formula)
Copy of AdS first law onto the cosmological horizon	**false**, Thm. **??**
$\dim \mathfrak{so}(1,4) < \dim \mathfrak{so}(2,4)$	[**P**], audited
CHM net of balls from dS isometries	**false**, Thm. **??**
FGHMV-standard closure of Q2	**obstructed**
Einstein+Λ from a cosmological CFT	still not derived; not [**P**]
Jacobson route	still [**C**]; not used
AdS results of the series	unchanged, [**P**]

Admission. I did not derive Einstein–Cartan in de Sitter. I proved that the standard this program uses for a gravitational [**P**] cannot be copied onto the cosmological horizon: the first-law sign is opposite and the isometries are too few. That is the [**P**]. A dual was not invented. Hehl–Datta was not recovered on a horizon.

The phrase that is true: *FGHMV-standard de Sitter closure is obstructed.* The phrase that is false: *we have cosmological Einstein from entanglement.*

References

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